

TiHAN–IIT Hyderabad

DST NM-ICPS Technology Innovation Hub on Autonomous Navigation
and Data Acquisition Systems (UAVs, ROVs, etc.)

Design and Development of Efficient Road Object Detection Methods in Extreme Weather Conditions

Revised Execution Plan — Focused 13-Week Delivery Sprint

Parent Project	Design and Development of Efficient Road Object Detection Methods in Extreme Weather Conditions for Secure Autonomous Driving (TiHAN Proposal)
Principal Investigator	Dr. Chalavadi Vishnu, Assistant Professor, CSE, IIT Tirupati
Co-Principal Investigator	Dr. Yeturu Kalidas, IIT Tirupati
Executing Team	3 student members, part-time
Planning Window	17 August 2026 – 15 November 2026 (13 weeks)
Target Venue	IEEE Intelligent Vehicles Symposium (IV) 2027 — submission deadline 15 Nov 2026
Document Date	17 August 2026

This document is a supervisor-facing companion to the internal team execution plan. It presents a deliberately scoped-down, time-boxed plan for advancing the parent TiHAN proposal's objectives within the team's actual available effort, and identifies a concrete target publication venue and deadline.

1. Executive Summary

The parent proposal (Annexure A, submitted 06 March 2025) scopes an 8-month effort across four stages: extreme-weather understanding, weather-robust object detection, edge-device optimisation, and adversarial robustness. Rather than attempting all four stages in parallel, this plan concentrates the available effort on one coherent, finishable contribution that builds directly on work already completed: an Attention-Gated Pix2Pix restoration model for smoke removal (PSNR 25.30 dB / SSIM 0.79, versus 21.45 dB / 0.72 for vanilla Pix2Pix). The plan extends this to two additional weather conditions, evaluates restoration quality using downstream detection accuracy rather than image-quality metrics alone, and closes out with real measurements on embedded hardware.

The plan targets submission to the IEEE Intelligent Vehicles Symposium (IV) 2027 (paper deadline 15 November 2026), with IEEE ITSC 2027 and a national-level venue identified as fallback options. Items outside this scope — diffusion-based restoration, transformer/foundation-model detection, adversarial attack generation and defence, and domain adaptation — are retained as an explicit Future Work roadmap (Section 12) rather than dropped from the project.

2. Background and Current Status

Advanced Driver Assistance Systems (ADAS) rely on vision and LiDAR perception that degrades sharply in fog, rain, snow, and smoke. The parent proposal identifies four linked problems: (i) scarcity of labelled extreme-weather driving data, (ii) degraded object-detection accuracy under adverse conditions, (iii) the computational cost of the architectures capable of handling these conditions, and (iv) the vulnerability of such systems to adversarial and naturally occurring noise.

Work completed to date:

- An Attention-Gated Pix2Pix conditional GAN (U-Net generator, PatchGAN discriminator, adversarial + $\lambda \cdot L1$ loss) trained for smoke removal on a Kaggle smoke-image dataset, 100 epochs.
- Quantitative result: PSNR 25.30 dB, SSIM 0.79, versus 21.45 dB / 0.72 for a vanilla Pix2Pix baseline — a clear, reproducible improvement from the attention mechanism.
- Target deployment hardware (NVIDIA Jetson / edge accelerator) and the next steps — multi-weather generalisation, INT8 quantisation, and robustness analysis — already identified in the team's own roadmap.

3. Revised Scope and Rationale

This plan scopes the work to what can be finished, evaluated, and written up to publication quality inside the available time, while preserving the cut items as a clearly defined next phase.

Retained This Cycle	Deferred to Future Work (Section 12)
Restoration extended to 2 additional weather types (rain, fog/snow), reusing the validated attention-gated architecture	Diffusion-model-based restoration
One efficient detector (YOLOv8n/s) fine-tuned and evaluated on weather data	Full DETR / ViT / foundation-model detector comparison
Restoration → detection pipeline evaluated with task-driven metrics (mAP), not image quality alone	Adversarial attack generation and defensive-distillation study
ONNX → TensorRT export, FP16/INT8 quantisation, measured FPS / latency / power on real edge hardware	Unsupervised domain adaptation across sensor domains
One conference-quality submission (IEEE IV 2027)	New ground-truth data collection/annotation

If the team completes the retained scope ahead of schedule, effort shifts to the deferred items in the order listed, beginning with the adversarial robustness study, which is the most natural continuation for a second paper.

4. Objectives

- Extend the existing attention-gated restoration model to rain and fog/snow degradations, in addition to the already-completed smoke condition.
- Quantify the effect of restoration on downstream object-detection accuracy (mAP@50, mAP@50:95) across clean, degraded, and restored conditions — establishing that restoration measurably improves perception, not only image-quality scores.
- Export and quantise the pipeline (FP16 and INT8) and report real latency, frame rate, and power draw on embedded hardware, directly addressing the team's own flagged gap of 12 FPS against a >30 FPS target.
- Produce one submission-ready paper for IEEE IV 2027, with a national-venue submission as a secondary, lower-risk output if time permits.

5. Proposed Methodology

5.1 Multi-Weather Image Restoration

The validated Attention-Gated Pix2Pix architecture is retained rather than redesigned, to control risk. Two additional single-weather models (rain, fog/snow) are trained first, using the existing training pipeline, to guarantee a working deliverable early in the schedule. If capacity remains, a stretch extension merges these into a single conditional generator with a weather-type embedding, which would reduce the number of model weights that must later be deployed on the edge device — a meaningfully stronger result for the paper if it converges reliably.

5.2 Weather-Robust Object Detection

YOLOv8n or YOLOv8s (Ultralytics) is used for detection, chosen for its edge-deployability and direct ONNX/TensorRT export path. The detector is fine-tuned on weather-tagged driving data and evaluated on the DAWN dataset — real road photographs in fog, rain, snow, and sandstorm, annotated with vehicle and pedestrian bounding boxes, already cited in the parent proposal. For each weather type, detection accuracy is compared across three conditions: clean baseline, degraded (no restoration), and restored-then-detected. This three-way comparison is the paper's central evidence that restoration helps perception, and is a stronger claim than PSNR/SSIM improvement alone.

5.3 Edge Deployment and Optimisation

The restoration and detection models are exported through ONNX to TensorRT and benchmarked at FP16 and INT8 precision on an NVIDIA Jetson-class device. Frame rate, latency, memory footprint, and power draw are measured on-device (via tegrastats), and any accuracy loss from quantisation is reported alongside the speed gain rather than omitted.

5.4 Evaluation Methodology

- **Primary metric:** detection mAP@50 and mAP@50:95, per weather type, across clean / degraded / restored conditions.
- **Secondary metric:** restoration PSNR/SSIM, reported alongside mAP to make the task-driven-vs-image-quality distinction explicit.

- **Efficiency metric:** latency–accuracy trade-off (FPS vs. mAP) across FP32 / FP16 / INT8, presented as a Pareto-front figure.
- **Qualitative analysis:** representative success and failure cases, including at least one honestly reported failure mode (e.g. heavy snow occlusion).

6. Datasets

Purpose	Dataset(s)	Notes
Restoration training — smoke	Existing Kaggle smoke dataset	Already in use; validated baseline
Restoration training — rain	Rain100H / Rain100L, RainCityscapes	Standard paired-rain benchmarks
Restoration training — fog/haze	RESIDE (SOTS / ITS / OTS)	Standard paired-haze benchmark
Restoration training — snow	Snow100K / CSD	Standard paired-snow benchmark
Primary joint restoration+detection evaluation	DAWN dataset	Real road scenes, fog/rain/snow/sandstorm, with detection bounding boxes; already cited in the parent proposal
Detector fine-tuning at scale	BDD100K (weather subset), Foggy Cityscapes	Larger fine-tuning pool; DAWN held out as the test set
Optional qualitative check	ACDC dataset	Real adverse-condition imagery with correspondences

New ground-truth data collection is out of scope for this cycle; existing labelled datasets are used throughout, with synthetic augmentation (Albumentations weather transforms) as a fallback only where a real dataset is too small.

7. Tools, Frameworks, and Hardware Requirements

7.1 Software

Category	Tool
Deep learning framework	PyTorch, torchvision
Detection	Ultralytics YOLOv8
Model export	ONNX, onnxruntime
Edge inference	NVIDIA TensorRT, JetPack SDK
Weather augmentation	Albumentations
Differentiable image ops	Kornia (for stretch-goal conditional model)
Experiment tracking	Weights & Biases (or TensorBoard)
Version control	Git / GitHub, single shared repository
Data/model versioning	DVC or a single shared, consistently named storage location
Manuscript preparation	Overleaf, official IEEE conference LaTeX template

7.1a Pre-Hardware Simulation and Benchmarking Tools

Before committing to Jetson deployment, the pipeline should be validated and profiled on the training machine using simulation-level tooling. This catches export errors early and provides a relative FP32/FP16/INT8 baseline without occupying the physical edge device.

Ultralytics Built-in Benchmark Mode (recommended first step)

Since the project already uses YOLOv8, the Ultralytics *benchmark* utility exports the model across formats (PyTorch, ONNX, TensorRT, TFLite, etc.) and reports speed, FPS, and mAP together in one table — mapping directly onto the Pareto-front requirement (FP32 vs FP16 vs INT8, FPS vs mAP):

```
from ultralytics.utils.benchmarks import benchmark

benchmark(
    model='yolov8n.pt',
    data='your_dataset.yaml',
    imgsz=640,
    half=False,          # set True for FP16
    device=0             # GPU index
)
```

There is also a *ProfileModels* class specifically for profiling ONNX and TensorRT builds side-by-side, reporting latency and FLOPs — useful for the restoration model, not just the YOLO detector. Running this on any CUDA GPU (training machine, Colab) will not produce Jetson-exact numbers, but it validates the PyTorch → ONNX → TensorRT export pipeline and gives a reliable relative sense of the FP16/INT8 speedup before the physical device is touched.

Other useful pre-hardware tools

- **ONNX Runtime Profiler** — run *onnxruntime.InferenceSession* with profiling enabled on your GPU; gives operator-level latency breakdown for the restoration model before TensorRT conversion.
- **torch.profiler / torch.utils.benchmark** — PyTorch-native profiling for the restoration GAN; useful for identifying bottleneck ops before export.
- **Jetson Power Estimator (NVIDIA web tool)** — spreadsheet-based power estimation for Jetson Orin Nano workloads; useful for sanity-checking power budget before hardware access is confirmed.
- **trtexec (TensorRT CLI)** — ships with TensorRT/JetPack; runs directly on the training machine if TensorRT is installed, benchmarking FP32/FP16/INT8 engine throughput without needing a Jetson.

Recommended workflow: Ultralytics benchmark → trtexec on training machine → on-device tegrastats on Jetson. The first two steps de-risk the export pipeline and establish a relative baseline; the third provides the real-world numbers for the paper.

7.2 Hardware

- **Training compute:** a GPU with ≥16–24 GB VRAM (e.g. RTX 3090/4090, A100) or equivalent cloud access (Google Colab Pro+, Kaggle Notebooks) for restoration and detector fine-tuning.
- **Edge benchmarking:** an NVIDIA Jetson Orin Nano (8 GB) development kit is recommended over the older, end-of-life Jetson Nano, to make the >30 FPS target realistically and honestly achievable. The parent proposal's own budget already includes a Jetson/RPi line item — availability should be confirmed with the PI before procurement.
- **Storage:** approximately 200–500 GB for the combined dataset set.

Confirming GPU cluster/cloud access and Jetson availability is the first action item in Week 0 — it is the single largest risk to the timeline below.

8. Execution Timeline

13 weeks, aligned to the IEEE IV 2027 deadline of 15 November 2026.

Weeks	Dates	Focus	Deliverable
0	17–23 Aug	Kickoff: confirm GPU/Jetson access; repo and experiment-tracking setup; reproduce existing smoke model; download datasets	Working repository; confirmed compute; reproduced baseline
1–2	24 Aug–6 Sep	Data pipeline; zero-shot detection baseline on clean vs. weather images	Motivating result: quantified detection degradation under weather
3–4	7–20 Sep	Extend restoration model to rain and fog/snow	Two additional restoration models, PSNR/SSIM table
5–6	21 Sep–4 Oct	Fine-tune detector; run three-way (clean/degraded/restored) evaluation on DAWN	Core results table: mAP by weather type and condition
7–8	5–18 Oct	Ablations (attention on/off, loss variants); stretch: unified conditional model	Ablation table; optional unified-model checkpoint
9–10	19 Oct–1 Nov	Edge export and quantisation; on-device FPS/latency/power benchmarking	Latency–accuracy Pareto table and figure, on real hardware
11	2–8 Nov	Consolidate figures/tables; core writing push	Near-complete manuscript draft
12	9–15 Nov	Proofread, format, PI review, submit	Submission to IEEE IV 2027

The team's availability extends roughly two weeks beyond the 15 November deadline; this margin is held as contingency rather than pre-allocated, and — if unused — redirected toward the fallback submission identified in Section 10.

9. Team Roles and Resource Allocation

Role	Primary Ownership	Weeks
Restoration	Multi-weather model training and ablations	1–4, 7–8
Detection & Evaluation	Zero-shot baseline, fine-tuning, three-way mAP evaluation	1–2, 5–8
Infrastructure & Deployment	Repository/tracking setup, ONNX/TensorRT export, edge benchmarking	0, 9–10

All three members contribute to manuscript writing throughout, rather than only in the final two weeks. A short weekly synchronisation (counted within the weekly allocation) is treated as mandatory: with three part-time contributors, coordination risk is larger than any single technical risk in this plan.

10. Target Publication Venues

Venue	Deadline	Role
IEEE Intelligent Vehicles Symposium (IV) 2027	15 Nov 2026 (paper); notification 15 Jan 2027	Primary target — direct topical fit, deadline fits the planning window
IEEE IV 2027 Workshop track	≈ 1 Feb 2027	Fallback if the main track does not accept
IEEE ITSC 2027	Expected ≈ March 2027 (based on ITSC 2026 pattern)	Secondary target with more runway; natural venue for the deferred Stage 3 work

National Conference on Computer Vision, Pattern Recognition, Image Processing and Graphics (NCVPRIPG), LNMIIT Jaipur	To be confirmed — verify directly with the organisers	Lower-barrier national venue for a second, easier submission
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Several other venues (WACV 2027, ITSC 2026, ICPR 2026, CVIP 2026) were reviewed and excluded: their deadlines have either already passed or fall within days of the current planning date, leaving insufficient time to produce the extended results this plan describes. A preprint may be posted to arXiv once the paper is submitted, which is permitted under IEEE conference policy and establishes priority while the submission is under review.

11. Risk Management

Risk	Mitigation
Compute or Jetson hardware access not confirmed in time	Resolved as the first Week 0 action item, before modelling work begins
Conditional multi-weather model fails to converge	Treated as a stretch goal only; per-weather models are the guaranteed fallback
INT8 quantisation causes unacceptable accuracy loss	FP16 reported as the primary deployment result; INT8 reported honestly as an additional speed/accuracy trade-off point
Coordination drift across three part-time contributors	Mandatory weekly sync; shared experiment tracker; single repository
Manuscript writing compressed into the final two weeks	Manuscript drafted incrementally from Week 0 as results become available
15 November deadline missed for any reason	IEEE IV workshop track and IEEE ITSC 2027 retained as live fallback venues for the same material

12. Future Work (Post-Submission Roadmap)

The items deferred in Section 3 remain part of the parent proposal's objectives and are the natural next phase once the current submission is complete. With restoration, detection, and edge-deployment infrastructure already in place, this next phase is an extension rather than a rebuild:

- **Adversarial robustness:** attack generation (GAN/diffusion-based) and defence (adversarial training, defensive distillation) — targeted at IEEE ITSC 2027.
- **Broader architecture exploration:** diffusion-based restoration, DETR/ViT-based detection, and comparison against foundation-model backbones.
- **Unsupervised domain adaptation** across sensor domains and geographies.
- **New data collection** tailored to gaps identified during the current evaluation.

13. Conclusion

This plan converts the parent proposal's broader four-stage roadmap into a single, scoped, and finishable 13-week deliverable that builds directly on the team's existing validated result. It is designed to produce a genuine submission — not a partial prototype — to a well-matched, currently open venue, while explicitly preserving the remaining proposal objectives as a funded next phase.

Prepared by (Project Team)

Name: _____

Reviewed and Approved by (Supervisor)

Dr. Chalavadi Vishnu / Dr. Yeturu Kalidas

Date: _____

Date: _____